AI SKILL TREE 2026

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Want to get better at AI, but don't know what you don't know? From daily chat (L1) to autonomous coding agents (L3–L4) and IMO-math / cyber-RL (L6), here is the full progression map.

① DIAGNOSE: Scan down each column (L1 → L6) to find where you stop
② LEVEL UP: Target the Top 3 pills in the row below
③ UNLEARN: Avoid the OBSOLETE traps on the right

*Scope: Modern LLMs, Multimodal (mmLLM) & Agentic AI across end-users, MLE & model research (not GOFAI or classic ML/DL). Changes fast — see the AI Ladder Evolution mega-chart by Arian Prabowo · Updated: 2026-11-05 22:05:24 AEDT (UTC+11:00)

AI LEVEL	THINK, SEARCH & ANALYZE	CREATE & DESIGN (MEDIA)	AUTOMATE & BUILD (CODE)	OBSOLETE
Overview · Persona · Surface · Use Case	Research · Data & Docs · RAG · Reasoning & Alignment	Image · Video · Audio · Voice · 3D · Provenance & DiT	Workflows · AI IDEs · Agents · Evals, Safety & Training	Still Doing This? You're Behind
LEVEL 1 ▰▱▱▱▱▱ **EVERYDAY CONSUMER**	**ChatGPT / Claude Daily Q&A**	**ChatGPT Live Voice Mode**	— NO AUTOMATION / CODING AT LEVEL 1 —	**"Act as a..." Jailbreaks**
Uses default consumer AI apps as a conversational search engine, writing assistant, and instant media helper. • Persona: Casual daily user & general public. • Surface: Phone app, web browser & built-in OS assistant on any phone or laptop. • Use Case: Search replacement, drafting emails/texts, summarizing PDFs/videos, live voice & casual image gen.	**Perplexity Cited Web Search**	**Conversational Image Gen (4o)**	Everyday consumers use pre-built apps only. No-code workflows & canvases start at Level 2 ↓	**Buying 10k Prompt PDFs**
	PDF & Article Upload Summary	**Phone Camera Live Vision Q&A**		**Blindly Trusting AI Claims**
	Google Search AI Overviews · Gemini Mobile Assistant	Gemini Live Voice Chat · Google Lens Visual Q&A		ChatGPT w/o Web Search · Copy-Pasting 1 Para Only
	Web Browsing in Chat · Android Circle to Search	Mobile OS Voice Dictation · Magic Eraser Photo Unblur		ChatGPT Plugins Store · Google Bard / Bing Sydney
	Apple AI Writing Tools · Meta AI in WhatsApp / IG	Canva Magic Media Studio · iOS Genmoji / Image Wand		DALL-E mini / Craiyon · Short 3-Word Prompts
	Hallucination Fact-Check · Draft & Rewrite Emails	Screenshot OCR Translate · Zoom / Meet AI Summaries		Passwords in Public Chat · AI Voice-Clone Scams
	DeepSeek Free Web / App · TLDR Thread Summarize · Grammar & Tone Polish	Google Imagen / ImageFX · Bing Image Creator · Auto-Captions on Video		AI as Medical/Legal Fact · Ignoring Source Links
	YouTube Video Summary · Gmail / Outlook Auto-Draft · Foreign Language Chat	Spotify AI Playlist DJ · 1-Click BG Removal · Receipt Photo Expense		
	Travel Itinerary Planner · Excel / Sheets Formulas · Click Cited Source Links	Voice Memo Transcript · Google Photos Ask-Photos · Whiteboard Photo Read		
	Memory On / Off Toggle · Data Training Opt-Out · Temporary Incognito Chat	Menu Photo Translation · Spot AI / Deepfake Tells · Short-Video Auto-Clip		
LEVEL 2 ▰▰▱▱▱▱ **WORKSPACE POWER USER**	**Multi-Step Deep Research**	**Midjourney --sref / --cref Refs**	**Claude Artifacts (React / SVG)**	**"Let's think step by step"**
Orchestrates paid reasoning models, deep-research agents, and interactive canvases across multi-file workspaces. • Persona: Knowledge worker, analyst, designer & solo operator. • Surface: Paid web workspaces, desktop apps, multi-source notebooks & creative studios. • Use Case: Deep research reports, multi-doc synthesis, interactive canvases & consistent media.	**Extended Thinking (o3 / Claude)**	**Sora 2 / Veo 2 / Runway Video**	**ChatGPT Agent / Operator Mode**	**"Prompt Engineer" ($300k)**
	Data Analysis & Chart Canvas	**ElevenLabs Instant Voice Clone**	**Zapier / Make No-Code AI Flows**	**Uploading Client PII to Chat**
	Projects & Custom GPTs · NotebookLM 50-Source Pod	NotebookLM Audio Host · Kling / Luma Keyframes	ChatGPT Canvas Co-Editor · Scheduled AI Tasks & Alerts	Tipping $200 / Threats · "6 fingers" Neg Prompts
	DeepSeek-R1 DeepThink · Gemini 1M–2M Token Upload	Ideogram / 4o Text-Image · Suno v4 / Udio Song Gen	Notion AI & M365 Copilot · Raycast AI Quick Commands	Midjourney Discord Bot · Custom GPT Store Hype
	Perplexity Pro Spaces · Multi-Doc Cross-Audit	HeyGen Video Translate · Superwhisper / Wispr Flow	Granola / Fireflies Notes · Airtable / Sheets AI Agents	Manual 4k Text Splits · 1-Page Prompt Walls
	Zero-Data-Retention (ZDR) · Critique-Then-Revise Loop	Adobe Firefly Gen-Fill · Live Screen-Share Vision	Custom GPT Action Webhooks · Workspace SSO & Audit Logs	Copy-Pasting CSV Tables · Manual Zoom Note-Taking
	Grok-3 DeepSearch Mode · LMSYS Chatbot Arena · Side-by-Side Model Check	Pika Scene Ingredients · Descript Overdub & Eye · Topaz AI Video Upscale	Apple Shortcuts AI Actions · Browser AI Sidebar Actions · Linear / Jira AI Triage	Fake Legal Citations · Shadow AI w/o IT SSO
	Context Token Trimming · CSV / Excel Code Sandbox · PII Redaction Before Upload	Captions.ai Avatar · Hedra Character-2 Sync · Leonardo Phoenix Canvas	Superhuman / Gmail AI Rules · Slack AI Channel Digests · Loom AI Auto-Workflows	
	Full-Book PDF Q&A · Earnings Call Synthesis · Literature Review Matrix	Recraft Brand Vectors · Spline AI 3D Prompts · Stem Splitter Isolation	Tally / Typeform AI Routing · Coda AI Doc Automations · TextExpander / Alfred Snips	
	Rubric Self-Grading · Role-Separated Debate · Citation Source Audit	Photoshop Gen-Expand · Commercial IP-Safe Models · Voice Consent Verification	Readwise Reader AI Ghost · Mermaid / Whimsical AI Flow · Role-Based Access (RBAC)	
LEVEL 3 ▰▰▰▱▱▱ **VIBE CODER & BUILDER**	**Spec-First PRD & XML Prompts**	**ComfyUI Ready Workflows**	**Cursor / Windsurf Agent Mode**	**AutoGPT / BabyAGI Loops**
Ships full-stack apps and multi-step automations by directing AI IDEs, CLI coding agents, and cloud APIs. • Persona: Indie builder, automation maker & AI-native developer. • Surface: AI IDEs (Cursor/Windsurf), CLI agents (Claude Code), app builders & APIs. • Use Case: Natural-language app shipping, multi-file repo refactoring, MCP/webhook flows & API media pipes.	**Tavily / Exa / Firecrawl API**	**Fal.ai / Replicate Webhooks**	**Claude Code CLI & CLAUDE.md**	**Ghost-Text Autocomplete**
	Repomix / Gitingest Packing	**ElevenLabs / Vapi Voice API**	**Lovable / Bolt / v0 / Replit**	**Blindly Running Agent rm -rf**
	System vs User Role Split · llms.txt Docs Standard	Black Forest Flux API · Remotion Code-to-Video	.cursor/rules & AGENTS.md · Claude Desktop MCP & n8n	API Keys in Public Git · Copy-Pasting StackTraces
	Jina Reader URL-Markdown · Context Window Budgeting	Whisper + FFmpeg Subs · Sync.so Lip-Sync API	Context7 Live Docs MCP · Plan-Before-Act & Diffs	Legacy .cursorrules Dump · 50-Node Flowise Chains
	Markdown Memory Banks · Multi-Step Prompt Chains	Recraft v3 Vector API · Figma-to-Code Plugins	Human-in-Loop Tool Approve · Cline / Roo Code / Aider	Manual Regex Scraping · Pasting Files to Web UI
	Few-Shot JSON Examples · OpenRouter Benchmarking	Magnific / Krea Upscale · Luma / Meshy 3D Assets	Browser-Use / Stagehand · Vercel AI SDK & Supabase	No Git Commit Pre-Agent · Shipping AI SQL w/o Auth
	Perplexity Sonar API · Brave Search Grounding · Unstructured PDF API	Runway Act-One API · Hume Empathic Voice API · Retell Phone Webhooks	Ollama / LM Studio Local · Groq / Cerebras LPU API · LiteLLM / CF AI Gateway	No API Spend / Rate Cap · Fake Slopsquat Packages
	Prompt Git Versioning · Token Cost & Spend Caps · Fallback Model Retries	AssemblyAI Diarization · Creatomate JSON Video · Photoroom Batch API	Together / Fireworks API · Helicone / LangSmith Logs · AI Code Security Scan	
	Streaming SSE Responses · Temp / Top-P API Tuning · Stop-Sequence Control	LivePortrait Face API · Mux Video Auto-Chapters · PlayHT Low-Latency TTS	Playwright AI E2E Tests · Secret & API-Key Scrubbing · Regex / Bash / Git via AI	
	Needle-in-Haystack Check · Untrusted Input Delimiters · SQL / Pandas Query via AI	Gladia Multilingual ASR · C2PA / SynthID Metadata · NSFW Moderation API Gate	GitHub Actions PR Bot · Slopsquat Package Check · Stripe + Auth0 Vibe Ship	
LEVEL 4 ▰▰▰▰▱▱ **AI & AGENT ENGINEER**	**Pydantic / Zod Schema Outputs**	**OpenAI / Gemini WebRTC API**	**Custom MCP Servers & OAuth**	**Shipping w/o Evals or Traces**
Architects production multi-agent backends with schema-enforced tool use, hybrid RAG, guardrails, audit traces, and CI evals. • Persona: Production AI software engineer & systems architect. • Surface: Python/TS SDKs, custom MCP servers, state graphs, microVM sandboxes & observability stacks. • Use Case: Stateful multi-agent backends, hybrid RAG, guardrails, audit trails, compliance & CI eval gates.	**Hybrid RAG + Cross-Rerankers**	**LiveKit / Pipecat Voice AI**	**LangGraph Durable State Graphs**	**Naive 500-Tok Vector RAG**
	ColPali & Docling PDF RAG	**Programmatic ComfyUI API**	**Promptfoo / Braintrust CI Evals**	**Regex-Parsing LLM JSON**
	Prompt Caching (90% Saved) · Hallucination & Faithfulness	Deepgram Nova-3 Stream · Cartesia / Kokoro TTS	E2B / Modal Sandboxes · Audit Trails & OTel Traces	Un-Sandboxed Code Exec · Fine-Tuning to Add Facts
	Anthropic Thinking Tokens · Contextual & GraphRAG	Vision-LLM UI Grounding · Barge-In VAD Interruption	Llama-Guard / NeMo Rails · LLM-as-a-Judge Rubrics	LangChain LLMChain Class · Assistants API v1 Polls
	DSPy Prompt Optimization · BAML Typed Prompt Calls	Twilio SIP Call Bridging · TwelveLabs Video Search	Git Worktree Subagents · OpenAI SDK / PydanticAI	text-embedding-ada-002 · Pinecone Pod-Index Lock
	Cohere Rerank-v3.5 Pipe · Citation Span Grounding	CLIP / SigLIP Visual RAG · WebSocket PCM16 Streams	Qdrant / Milvus / pgvector · SWE-bench Eval Harness	No Injection Guardrails · No Tool Audit Logging
	Hypothetical Embed (HyDE) · Parent-Doc Late Chunking · Reciprocal Rank Fusion	Frame-Accurate Video Cut · Silero VAD Turn-Taking · Daily.co WebRTC Mesh	Prompt Injection Red-Team · EU AI Act / SOC2 Audit Logs · Firecracker MicroVMs	No Model Fallback Route · Vibe-Check Only Evals
	Corrective RAG (CRAG) · Semantic Intent Router · Multi-Query RAG Expand	HeyGen Streaming Avatar · WhisperX Word Timestamps · pyannote Speaker Diarize	Human-in-Loop Interrupts · Agent Memory (Mem0/Letta) · Temporal Durable Exec	
	Surya / LayoutLM Tables · Demographic Fairness Eval · PII & Bias Leakage Audits	Bounding-Box Detection · Deepfake / Spoof Detection · C2PA Cryptographic Sign	Redis Semantic Caching · Langfuse / Phoenix Spans · Cascade Model Cost Router	
	Faithfulness Gate Check · Time-Weighted Memory · Golden Retrieval Recall	Sub-500ms Voice Loop · Audio Watermark Check · 3D Gaussian Asset API	Canary / Shadow Deploys · RBAC Tool-Permission Gate · A2A Agent2Agent Protocol	
LEVEL 5 ▰▰▰▰▰▱ **LOCAL & POST-TRAINING**	**Synthetic CoT Dataset Curation**	**ComfyUI Custom Node Pipelines**	**vLLM / SGLang GPU Serving**	**Full-Weight 7B Style FT**
Serves quantized open-weight models on dedicated GPUs/Macs and aligns weights via QLoRA, DPO, GRPO, and steering. • Persona: ML systems engineer, local-first builder & model post-trainer. • Surface: Multi-GPU rig (RTX 3090/4090/5090), unified-memory Mac, vLLM, SGLang & Unsloth. • Use Case: High-throughput local serving, QLoRA/DPO/GRPO alignment, weight merging & air-gapped AI.	**Rejection-Sampled SFT / DPO**	**Flux.1 / Wan-2.1 Local Weights**	**Unsloth / Axolotl / TRL QLoRA**	**Automatic1111 SD 1.5 UI**
	Grammar-Constrained GBNF / JSON	**Kohya / AI-Toolkit LoRA Train**	**DPO / ORPO / Local GRPO Tune**	**SFT Without KL / Eval Checks**
	Local DeepSeek-R1 & Qwen-3 · Constitutional / RLAIF Data	ControlNet & IP-Adapter · Whisper-v3-turbo Local	EXL2 / AWQ / GGUF / FP8 · SGLang RadixAttention	8-bit bitsandbytes Slow · Oobabooga Text-Gen UI
	RoPE Scaling (YaRN) · Min-P / DRY / XTC Sample	Kokoro-82M / F5-TTS Local · RVC / So-VITS Voice Clone	Outlines / XGrammar FSM · Speculative Decode (EAGLE)	Manual GPTQ Calibration · DreamBooth 40m Colab
	DPO Chosen/Reject Pairs · Custom ChatML Templates	Florence-2 & SAM-2 Vision · GGUF FP8 Flux Offload	MergeKit SLERP/TIES/DARE · Activation Steering Vectors	LoRA Rank-128 Overfit · Raw Alpaca 52k SFT Set
	Air-Gapped Hybrid RAG · Domain Eval Benchmarks	TensorRT Diffusion Accel · HunyuanVideo / Mochi Run	KV-Cache FP8 / INT4 Quant · Apple MLX / Dual-4090 Rig	Static RoPE Alpha Hacks · Flask Single-Batch API
	Refusal Boundary Tuning · Bias & Toxicity Filtering · Perplexity Quant Audit	TeaCache / WaveSpeed · Chatterbox / Orpheus TTS · faster-whisper CTranslate	llama.cpp / KoboldCpp · LLaMA-Factory Training · vLLM Multi-LoRA Serving	Blind Frankenmerges · FT Without Safety Evals
	KL-Divergence Drift Check · De-Duplicated FT Corpora · ShareGPT Multi-Turn Data	AnimateDiff Motion Nodes · Depth-Anything ControlNet · InstantID / FaceID Nodes	Qwen-Coder-32B Local IDE · Machine Unlearning Runs · Local Stdio MCP & SQLite	
	System Prompt Distill · Local BGE-Reranker · Air-Gapped Citations	CosyVoice2 / Fish-Speech · Ultimate-SD Tiled VAE · SageAttention Speedup	Hermes-3 Local Tool Call · OmniParser GUI Grounding · Adversarial Jailbreak Eval	
	CoT Trace Length Tuning · RAGAS Local Faithfulness · Data Contamination Check	JoyCaption VLM Tagging · Concept Erasure LoRA · Watermark-Preserving Gen	Prefill-Decode Disagg · lm-eval-harness Local · imatrix Importance Quant	
LEVEL 6 ▰▰▰▰▰▰ **FRONTIER RESEARCHER**	**RLVR (Verifiable Reward RL)**	**Rectified Flow Matching (DiT)**	**FSDP2 / Megatron 3D Parallel**	**4-Model PPO RLHF Pipeline**
Pre-trains foundation models, designs verifiable RL & scalable-oversight alignment, and authors custom GPU kernels. • Persona: Foundation model scientist, alignment researcher & kernel author. • Surface: Multi-node H100/B200/TPU clusters, InfiniBand fabrics, FSDP2/Megatron & CUDA/Triton. • Use Case: Base/MoE pre-training, RLVR/GRPO, mechanistic interpretability, scalable oversight & GPU kernels.	**GRPO (Group Relative Policy)**	**Native Early-Fusion Omni-Modal**	**Custom Triton & FlashAttention-3**	**Dense-Only 500B+ Pretrain**
	Test-Time Compute Scaling	**3D Spatio-Temporal Video VAE**	**DeepSeek MLA & Aux-Free MoE**	**Unverifiable Code/Math RMs**
	Scalable Oversight & Debate · Reward Hacking Defense	Discrete Audio Tok (Mimi) · Consistency Flow Distill	Mech Interp & SAE Circuits · FP8 / MXFP8 Mixed Train	Scaling w/o Safety Evals · nn.DataParallel / DDP
	Process Reward Model (PRM) · MCTS Tree-Search Reason	Autoregressive Visual Tok · Interleaved Omni Attend	Exec-Guided Code RLVR · Zero-Day Cyber Fuzzing RL	Static Sinusoidal Pos · FP32 Unfused Attention
	AlphaProof / Lean-4 RL · Deliberative CoT Alignment	Chameleon / Transfusion · 3D Gaussian Splatting	ThunderKittens / CUTLASS · torch.compile & Inductor	GANs for Image / Video · Dropless MoE Aux-Loss
	Pass@k → Pass@1 Distill · Benchmark Decontam LSH	Full-Duplex Speech Model · JEPA / Genie-2 World Sim	InfiniBand & NCCL Tuning · OpenRLHF / VeRL Clusters	Human-Only RL Oversight · Contaminated Benchmarks
	Scheming & Sandbag Evals · Weak-to-Strong Generalize · Instruction Hierarchy RL	Cross-Modal Contrastive · Rotary 3D-RoPE Video Pos · CFG Guidance Distillation	ASL / CBRN Frontier Evals · Autonomous Replication Eval · Chinchilla / MuP Scaling	Unmonitored Latent CoT · Length-Biased RMs
	Aha-Moment Backtracking · VinePPO / ReST-MCTS* Tree · KL-Divergence Ref Clip	MMDiT Dual-Stream Blocks · VQ-GAN / FSQ Scalar Quant · MAR Masked Autoregressive	DualPipe Compute Overlap · Multi-Token Predict (MTP) · WSD Warmup-Stable-Decay	
	Curriculum Difficulty RL · Coconut Latent Space CoT · Math-Shepherd Verifiers	SoundStorm Parallel Audio · Latent Concept Erasure · Tree-Ring Latent Watermark	Ring-Attention 1M+ Context · FlexAttention & Pallas · NVSHMEM All-to-All MoE	
	CoT Monitorability Checks · Specification Gaming Fix · AlphaEvolve Search Loop	Spatio-Temporal Patchify · Distribution Match Distill · Physics Video Priors	Micro-Scaling NVFP4 · Zero-Bubble Pipe Parallel · Differential Privacy SGD	